

Timothy Chumley

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Education

Washington University in St. Louis

Ph.D. in Mathematics, 2013

Marquette University

M.S. in Mathematics, 2007

B.S. in Computer Science, Mathematics, 2005

Employment

Mount Holyoke College, Department of Mathematics & Statistics

Associate Professor on the John Stewart Kennedy Foundation, 2023 – present

Associate Professor, 2022 – present

Assistant Professor, 2016 – 2022

Iowa State University, Department of Mathematics

NSF Alliance for Building Faculty Diversity Postdoctoral Fellow, 2013 – 2016

Washington University in St. Louis, Department of Mathematics

Graduate Teaching Assistant and Instructor, 2008 – 2013

Teaching

Mount Holyoke College

Math 101: Calculus I (Spring 2019)

Math 102: Calculus II (Fall 2016, 2021, 2022, 2026; Spring 2025)

Math 203: Calculus III (Fall 2022, 2025; Spring 2023, 2026)

Math 206: Introduction to Proofs through Analysis (Fall 2024)

Math 211: Linear Algebra (Spring 2017)

Math 232: Discrete Math (Fall 2018)

Math 241: Dynamical Systems (Spring 2022, 2026)

Math 301: Real Analysis (Fall 2020, 2021; Spring 2023, 2025)

Math 333: Differential Equations (Spring 2018)

Math 339SP/Stat 344: Stochastic Processes (Spring 2017, 2018, 2019, 2022, 2024; Fall 2025)

Math 342: Probability (Fall 2016, 2017, 2018, 2024, 2026; Spring 2021, 2024)

Iowa State University

Math 166: Calculus II (Fall 2013, 2015)

Math 317: Theory of Linear Algebra (Spring 2014)

Math 501: Introduction to Real Analysis (Fall 2014)

Washington University

Math 1011: Introduction to Statistics (Summer 2009, 2010)

Math 155: Calculus I (Summer 2008)

Math 309: Matrix Algebra (Summer 2012)

Research Articles

1. T. Chumley, R. Feres, and Z. Zhang. Entropy production in knudsen thermodynamics of compartmented systems. *arXiv preprint*, 2026.
2. T. Chumley, M. Covey, C. Cox, and R. Feres. Chaotic lensed billiards. *Discrete Contin. Dyn. Syst.*, 45(9):3345–3374, 2025.
3. T. Chumley, R. Feres, L. A. Garcia German, and G. Yablonsky. Revisiting maxwell-smoluchowski theory: low surface roughness in straight channels. *Chem. Eng. Sci.*, 284:119477, 2024.
4. J. Ahmed, T. Chumley, S. Cook, C. Cox, H. Grant, N. Petela, B. Rothrock, and R. Xhafaj. Dynamics of the no-slip galton board. *arXiv preprint*, 2022.
5. T. Chumley, R. Feres, and M. Wallace. Exact discretization of harmonic tensors. *Potential Anal.*, 56(3):409–421, 2022.
6. T. Chumley, R. Feres, and L. A. Garcia German. Knudsen diffusivity in random billiards: spectrum, geometry, and computation. *SIAM J. Appl. Dyn. Syst.*, 20(3):1655–1682, 2021.
7. T. Chumley and R. Feres. Entropy production in random billiards. *Discrete Contin. Dyn. Syst.*, 41(3):1319–1346, 2021.
8. T. Chumley, S. Cook, C. Cox, and R. Feres. Rolling and no-slip bouncing in cylinders. *J. Geom. Mech.*, 12(1):53–84, 2020.
9. O. Aydogmus, T. Chumley, A. Matzavinos, and A. Roitershtein. Moran-type bounds for the fixation probability in a frequency-dependent Wright-Fisher model. *J. Math. Biol.*, 76(1):1–35, 2018.
10. T. Chumley, R. Feres, and H.-K. Zhang. Diffusivity in multiple scattering systems. *Trans. Amer. Math. Soc.*, 368(1):109–148, 2016.
11. T. Chumley, S. Cook, and R. Feres. From billiards to thermodynamics. *Comput. Math. Appl.*, 65(10):1596–1613, 2013.
12. T. Chumley. *Limit Theorems for Random Billiard Models*. PhD thesis, 2013.

Supervision and Mentoring

Doctoral dissertations — mathematics

Co-supervisor (with Renato Feres), Zhenyi Zhang, Washington University, 2027 (expected)

Committee member, Wentao Fan, Tufts University, 2026

Committee member, John Kaushagen, UMass Amherst, 2025

Co-supervisor (with Renato Feres), Maeve Covey, Washington University, 2022

Committee member, Bowei Zhao, Washington University, 2022

Co-supervisor (with Renato Feres), Luis Garcia German, Washington University, 2020

Undergraduate theses at MHC

Committee member, computer science major, 2026

Committee member, statistics major, 2025

Committee member, biochemistry major, 2023

Committee member, statistics major, 2022

Committee member, psychology major, 2022

Committee member, statistics major, 2021

Committee member, chemistry major, 2020

Committee member, chemistry major, 2019

Committee member, economics major, 2019

Committee member, biochemistry major, 2018

Committee member, psychology major, 2018

Committee member, computer science major, 2017

Undergraduate research at MHC

Co-supervisor (with Christopher Cox), 2 students, Spring 2025

Co-supervisor (With Christopher Cox), 8 students, Summer 2024

Co-supervisor (with Alanna Hoyer-Leitzel), 3 students, Summer 2023

Co-supervisor (With Christopher Cox), 2 students, Summer 2023

Co-supervisor (with Alanna Hoyer-Leitzel), 1 student, Spring 2023

Co-supervisor (with Christopher Cox), 1 student, Fall 2022

Supervisor, 2 students, Fall 2021

Supervisor, 3 students, Summer 2021

Supervisor, 1 student, Spring 2021

Supervisor, 1 student, Spring 2019

Supervisor, 1 student, Fall 2018

Supervisor, 3 students, Fall 2017

Supervisor, 2 students, Summer 2017

Presentations

Invited

Amherst College, Colloquium, April 2026

Joint Mathematics meetings, AIM Special Session on Little School Dynamics: Cool Dynamics Research by Researchers at PUIs, January 2023

AMS Eastern Sectional Meeting, Special Session on Connections Between Theoretical and Applied Dynamical Systems: A Session in Honor of the 60th Birthdays of Renato Feres and Boris Hasselblatt, October 2022

Tarleton State University, Colloquium, September 2019

SIAM Conference on Applications of Dynamical systems, Minisymposium on Thermodynamic Laws from Nonequilibrium Dynamics, May 2019

AMS Eastern Sectional Meeting, Special Session on Billiard Dynamics: Standard and Alternative Collision Models, September 2018

Joint Mathematics Meetings, Special session on Research by Postdocs of the Alliance for Diversity in Mathematics, January 2018

AIMS Conference on Dynamical Systems, Differential Equations and Applications, Special Session on Dynamical systems and their applications, July 2016

Washington University, Geometry seminar, April 2016

Mount Holyoke College, Colloquium, February 2016

Joint Mathematics Meetings, Special session on Research by Postdocs of the Alliance for Diversity in Mathematics, January 2016

Contributed

International Congress of Mathematicians, Probability session, August 2018

Seminar on Stochastic Processes, University of Virginia, March 2017

Professional Service

Chair, Prize Committee for MAA Halmos-Ford Prize, 2025 – 2026

Panelist for Pure Math program, Division of Mathematical Sciences, NSF, 2024

Member, Lathisms Scholarship Selection Committee, 2024

Member, Steering Committee for the Hudson River Undergraduate Mathematics Conference, 2023 – 2025

Member, Prize Committee for MAA Halmos-Ford Prize, 2023 – 2026

Co-organizer for Special Session at AMS Fall Eastern Sectional Meeting, 2022

Panelist for Applied Math program, Division of Mathematical Sciences, NSF, 2022

Co-organizer, BIRS Oaxaca workshop on Frontiers in Billiard Dynamics, June 2021

Co-organizer, SIAM Conference on Applications of Dynamical systems mini-symposium, May 2019

Co-organizer for Special Session at AMS Fall Eastern Sectional Meeting, September 2018

Judge for MAA Undergraduate Poster Session at JMM, January 2017

Co-organizer for Special Session at 11th AIMS Conference on Dynamical Systems Differential Equations and Applications, Orlando, July 2016

Co-organizer for AIM SQuaRE on Stochastic thermodynamics and random billiards, 2015 – 2017

Co-organizer, Iowa State University probability seminar, 2014 – 2016

Co-organizer for AMS Special session on Statistical Aspects of Dynamical Systems, October 2013

Referee for Chaos: An Interdisciplinary Journal of Nonlinear Science, Discrete and Continuous Dynamical Systems, Journal of Mathematical Physics

Awards, Grants, and Other Honors

Interviewed for AMS Mathematical Moments, 2022

Featured in Latinxs and Hispanics in the Mathematical Sciences AMS poster, 2022
 Featured in MAA Focus article on Latinxs and Hispanics in the Mathematical Sciences, 2020
 Latinxs and Hispanics in the Mathematical Sciences (Lathisms) honoree, 2020
 AMS-NSF International Congress of Mathematicians travel award, 2014, 2018, 2022
 AMS Project NExT Fellow, 2016 – 2017
 American Institute of Mathematics SQuaRE grant, 2015 – 2017
 NSF Alliance for Diversity in Mathematical Sciences Postdoctoral Fellowship, 2013 – 2016
 Dissertation Fellowship, Washington University, 2012 – 2013
 University Fellowship, Washington University, 2007 – 2008, 2010
 Phi Beta Kappa, 2005

Selected Conference Participation

Resilience across dynamical systems in a stochastic world, BIRS, September 2026.
 Joint Mathematics Meetings, Boston, January 2023.
 AMS Eastern Sectional Meeting, Amherst, October 2022.
 Northeast Conference on Dynamical Systems, University of Massachusetts Amherst, November 2019.
 Illustrating Probability and Dynamics workshop, ICERM, November 2019.
 SIAM Conference on Applications of Dynamical systems, Snowbird, May 2019.
 AMS Eastern Sectional Meeting, Newark, September 2018.
 International Congress of Mathematicians, Rio de Janeiro, August 2018.
 Seminar on Stochastic Processes, Brown University, May 2018.
 Joint Mathematics Meetings, San Diego, January 2018.
 Mathfest, Chicago, August 2017.
 AIM SQuaRE on Stochastic thermodynamics and random billiards, June 2017
 Seminar on Stochastic Processes, University of Virginia, March 2017.
 Joint Mathematics Meetings, Atlanta, January 2017.
 Mathfest, Columbus, August 2016.
 The 11th AIMS Conference on Dynamical Systems, Differential Equations and Applications, Orlando, July 2016.

Service at Mount Holyoke College

Ad-hoc Committee on the Status and Future of Lecturer and Senior Lecturer Positions, 2024 – 2025
 High Performance Computing Cluster Committee, 2023 – 2026
 Faculty Conference Committee, 2023 – 2026
 Academic Administrative Board member, 2020 – 2023
 Math/Stat liaison for STEMPOC student group, 2018 – 2020
 Subcommittee on the Goldwater Fellowship, 2017 – 2020
 Math/Stat TAs and graders coordinator, 2017 – 2018, 2025 – 2027

Data science curriculum committee, 2016 – 2017

Problem solving club co-advisor, 2016 – 2017

Math/Stat LITS liaison, 2016 – 2017

Professional Memberships

Member, American Mathematical Society, 2007 – Present.

Member, Mathematical Association of America, 2016 – Present.

Last updated: July 29, 2026